$\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$, . . .

f_x , f_{xx} , f_{xxx} , . . .

$f'(x)$, $f''(x)$, $f'''(x)$, . . .

y' , y'' , y''' , . . .

THE GENERAL RULE

Given : $y = x^n$

then $\frac{dy}{dx} = n x^{n-1} \frac{d(x)}{dx}$

Eg 1 If $y = x^8$, find $\frac{dy}{dx}$

Solu : $\frac{dy}{dx} = 8x^7 \frac{d}{dx}(x) = 8x^7(1)$

$= 8x^7$

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If n is a positive integer, then

$$\frac{d}{dx}(x^n) = n x^{n-1} \cdot \frac{d(x)}{dx}$$

1. $\frac{d}{dx}(x^5) = 5x^4$

2. $\frac{d}{dx}(x^{12}) = 12x^{11}$

3. $\frac{d}{dx}(x) = 1 x^0 = 1$

4. $\frac{d}{dx}(4x^8) = 4\frac{d}{dx}(x^8) = 4(8x^7) = 32x^7$

5. $\frac{d}{dx}(x^4 + 6x^{11}) = \frac{d}{dx}(x^4) - \frac{d}{dx}(6x^{11})$

$= 4x^3 - 6 \cdot 11x^{10}$

$= 4x^3 - 66x^{10}$

6. $\frac{d}{dx}(c) = 0$; $\frac{d}{dx}(12) = 0$

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Using the power rule to find derivatives of functions:

$$\frac{d}{dx}(x^n) = n x^{n-1} \frac{d(x)}{dx}$$, where x is a function

Given $f(u) = u^n$

Then $$\frac{d}{dx}(u^n) = \frac{d}{du}(u^n)\frac{du}{dx}$$

$$= n u^{n-1} \frac{du}{dx}$$

Example:

Find $\frac{d}{dx}(x^2-1)^{50}$

Soln: $$\frac{d}{dx}(x^2-1)^{50} = 50(x^2-1)^{49}\frac{d}{dx}(x^2-1)$$

$$= 50(x^2-1)(2x)$$

$$= 100x(x^2-1)$$

Example:

$\frac{d}{dx}(x^2-1)^4$

Soln: $$\frac{d}{dx}(x^2-1)^4 = 4(x^2-1)^3\frac{d}{dx}(x^2-1)$$

$$= 4(x^2-1)^3(2x)$$

$$= 8x(x^2-1)^3$$

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Ex 2. $y = x^{13}$

$$\frac{dy}{dx} = 13x^{12}\frac{d}{dx}(x)$$

$$= 13x^{12}(1)$$

$$= 13x^{12}$$

Ex 3 $y = (5x)^8$

$$\frac{dy}{dx} = 8(5x)^7\frac{d}{dx}(5x)$$

$$= 8(5x)^7(5)$$

$$= 40(5x)^7$$

$$= 40(5)^7x^7$$

$$=$$

Ex 4 $y = (x^2+4)^5$

$$\frac{dy}{dx} = 5(x^2+4)^4\frac{d}{dx}(x^2+4$$

$$= 5(x^2+4)^4(2x)$$

$$= 10x(x^2+4)^4$$

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5 $y = \sqrt[3]{(1+x^2)^4}$

Soln: $y = (1+x^2)^{4/3}$

$$\frac{dy}{dx} = \frac{4}{3}(1+x^2)^{\frac{1}{3}} \frac{d}{dx}(1+x^2)$$

$$= \frac{4}{3}(1+x^2)^{\frac{1}{3}}(2x)$$

$$= \frac{8}{3}x(1+x^2)^{\frac{1}{3}}$$

6 $y = \sqrt{2x+7}$

In: $y = (2x+7)^{\frac{1}{2}}$

$$\frac{dy}{dx} = \frac{1}{2}(2x+7)^{-\frac{1}{2}} \frac{d}{dx}(2x)$$

$$= \frac{1}{2}(2x+7)^{-\frac{1}{2}}(2)$$

$$= (2x+7)^{-\frac{1}{2}}$$

$$= \frac{1}{\sqrt{2x+7}}$$

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7 $y = (7+3x)^5$, $\frac{dy}{dx} = ?$

$$\frac{dy}{dx} = 5(7+3x)^4 \frac{d}{dx}(7+3x)$$

$$= 5(7+3x)^4(3)$$

$$= 15(7+3x)^4$$

Ex 8 $y = (2x-3)^{-2}$, $\frac{dy}{dx} = ?$

$$\frac{dy}{dx} = -2(2x-3)^{-3} \frac{d}{dx}(2x-3)$$

$$= -2(2x-3)^{-3}(2)$$

$$= -4(2x-3)^{-3}$$

$$= -\frac{4}{(2x-3)^3}$$

Ex 9. $y = (3x^2+5)^{-3}$

$$\frac{dy}{dx} = -3(3x^2+5)^{-4} \frac{d}{dx}(3x^2+5)$$

$$= -3(3x^2+5)^{-4}(6x)$$

$$= -\frac{18x}{(3x^2+5)^4}$$

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Ex 10 $y = (x^3 - 3x^2 + 7x - 3)^4$

$$\frac{dy}{dx} = 4(x^3 - 3x^2 + 7x - 3)^3 \frac{d}{dx}[x^3 - 3x^2 + 7x - 3]$$

$$= 4(x^3 - 3x^2 + 7x - 3)^3 (3x^2 - 6x + 7)$$

Ex 11 $f(x) = 7x^5 - 3x^4 + 6x^2 + 3x + 4$

$$f'(x) = 7(5)x^4 - 3(4)x^3 + 6(2)x + 3$$

$$= 35x^4 - 12x^3 + 12x + 3$$

Ex 12 $f(x) = 4x^6 + 2x^5 - 7x^2 + 2x + 5$

$$f_x = 24x^5 + 10x^4 - 14x + 2$$

$$f_{xx} = 24(5)x^4 + 10(4)x^3 - 14$$

$$= 120x^4 + 40x^3 - 14$$

$$f_{xxx} = 120(4)x^3 + 40(3)x^2$$

$$= 480x^3 + 120x^2$$

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Exx

1. $y = -8x^5 + \sqrt{3}x^3 - 7x$

$$\frac{dy}{dx} = -40x^4 + 3\sqrt{3}x^2 - 7$$

2. $y = -7x^6 + \sqrt{3}x^3 + 2\pi x^2 - 12$

$$\frac{dy}{dx} = -42x^5 + 3\sqrt{3}x^2 + 4\pi x$$

3. $y = 2x^{50} + 3x^{12} - 14x^2 + \sqrt[3]{7}x +$

$$y' = 100x^{49} + 36x^{11} - 28x + \sqrt[3]{7}$$

4. $y = x^{1/3}$

$$\frac{dy}{dx} = \frac{1}{3}x^{-\frac{2}{3}}$$

5. $y = (3x^2 + 5)^{-4}$

$$\frac{dy}{dx} = \frac{24x}{(3x^2+5)^2}$$

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LEVEL 2

Ex 1. If $f(x) = x^6 + \frac{1}{\sqrt[3]{x^2}}$, find $f'(x)$

Soln: $f(x) = x^6 + x^{-\frac{2}{3}}$

$f'(x) = 6x^5 + \left(-\frac{2}{3}\right)x^{-\frac{5}{3}}$

$= 6x^5 - \frac{2}{3}x^{-\frac{5}{3}}$

Ex 2. If $f(x) = 7x^3 + \frac{10}{\sqrt{x}}$, find $f'(9)$

Soln: $f(x) = 7x^3 + 10x^{-\frac{1}{2}}$

$f'(x) = 21x^2 + 10\left(-\frac{1}{2}x^{-\frac{3}{2}}\right)$

$= 21x^2 + \left(-5x^{-\frac{3}{2}}\right)$

$= 21x^2 - 5x^{-\frac{3}{2}}$

$f'(9) = 21(9^2) - 5(9)^{-\frac{3}{2}}$

$= 21(81) - \frac{5}{27}$

$= 1701 - \frac{5}{27}$

$= 1700.815$

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3. Find the derivative of $y = \frac{1}{(4x^2-3)^5}$

Soln: $y = \frac{1}{(4x^2-3)^5} = (4x^2-3)^{-5}$

$$\frac{dy}{dx} = -5(4x^2-3)^{-6} \frac{d}{dx}(4x^2-3)$$

$$= -5(4x^2-3)^{-6}(8x)$$

$$= -40x(4x^2-3)^{-6}$$

$$= -\frac{40x}{(4x^2-3)^6}$$

4. Find the derivative of $y = \sqrt[3]{3x-2}$

Soln: $y = \sqrt[3]{3x-2} = (3x-2)^{\frac{1}{3}}$

$$\frac{dy}{dx} = \frac{1}{3}(3x-2)^{-\frac{2}{3}} \frac{d}{dx}(3x-2)$$

$$= \frac{1}{3}(3x-2)^{-\frac{2}{3}}(3)$$

$$= (3x-2)^{-\frac{2}{3}}$$

$$= \frac{1}{\sqrt[3]{(3x-2)^2}}$$

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Find $\frac{dy}{dx}$ if $y = \frac{3}{\sqrt{x}} - 2\sqrt{x}$

Soln: $y = \frac{3}{\sqrt{x}} - 2\sqrt{x} = 3x^{-\frac{1}{2}} - 2x^{\frac{1}{2}}$

$$\frac{dy}{dx} = 3\left(-\tfrac{1}{2}\right)x^{-\frac{3}{2}} - 2\left(\tfrac{1}{2}\right)x^{-\frac{1}{2}}$$

$$= -\frac{3}{2}x^{-\frac{3}{2}} - x^{-\frac{1}{2}}$$

$$= \frac{-3}{2x^{3/2}} - \frac{1}{x^{\frac{1}{2}}}$$

$$= \frac{-3}{2\sqrt{x^3}} - \frac{1}{\sqrt{x}}$$

Find the derivative of $y = (2x^5 - 4x^3 - x)^3$

Soln: $y = (2x^5 - 4x^3 - x)^3$

$$\frac{dy}{dx} = 3(2x^5 - 4x^3 - x)^3 \frac{d}{dx}(2x^5 - 4x^3 - x)$$

$$= 3(2x^5 - 4x^3 - x)^3 (10x^4 - 12x^2 - 1)$$

$$= 3(10x^4 - 12x^2 - 1)(2x^5 - 4x^3 - x)^3$$

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Ex 7. If $y = 1000 + 8x - \frac{150}{x^2}$, find

Soln: $y = 1000 + 8x - 150x^{-2}$

$$\frac{dy}{dx} = 0 + 8 - 150(-2x^{-3})$$

$$= 8 + \frac{300}{x^3}$$

Ex 8. Differentiate $f(x) = \frac{5}{x^2} - \frac{6}{x}$ a

find $f'(\frac{1}{2})$

Soln: $f(x) = 5x^{-2} - 6x^{-1}$

$$f'(x) = -10x^{-3} + 6x^{-2}$$

$$= -\frac{10}{x^3} + \frac{6}{x^2}$$

$$f'(\tfrac{1}{2}) = -\frac{10}{(\frac{1}{2})^3} + \frac{6}{(\frac{1}{2})^2}$$

$$= -80 + 24$$

$$= -56$$

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Ex 9

If $y = \frac{4}{3x^2 - x + 5}$, find y'

$$y' = 4\frac{d}{dx}(3x^2 - x + 5)^{-1}$$

$$= 4(-1)(3x^2 - x + 5)^{-2}\frac{d}{dx}(3x^2 - x + 5)$$

$$= -4(3x^2 - x + 5)^{-2}(6x - 1)$$

$$= \frac{-4(6x - 1)}{(3x^2 - x + 5)^2} = \frac{4(1 - 6x)}{(3x^2 - x + 5)^2}$$

Ex 10.

Given $y = \sqrt[3]{(4x^2 + 3)^2}$, find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{d}{dx}(4x^2 + 3)^{\frac{2}{3}}$$

$$= \frac{2}{3}(4x^2 + 3)^{-\frac{1}{3}} \cdot \frac{d}{dx}(4x^2 + 3)$$

$$= \frac{2}{3}\left(\frac{1}{4x^2 + 3}\right)^{\frac{1}{3}}(8x)$$

$$= \frac{16x}{3(4x^2 + 3)^{\frac{1}{3}}}$$

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11 Find the derivative of $y = (6x^5 - 4x$

Soln: $y = (6x^5 - 4x^3 - 5)^7$

$\frac{dy}{dx} = 7(6x^5 - 4x^3 - 5)^6 \frac{d}{dx}(6x^5 - 4$

$= 7(6x^5 - 4x^3 - 5)(30x^4 - 12x$

$= 7(30x^4 - 12x^2)(6x^5 - 4x^3$

$= 7(6)(5x^4 - 2x^2)(6x^5 - 4x^3$

$= 42x^2(5x^2 - 2)(6x^5 - 4x^3$

12 Find the derivative of $y = \frac{2}{\sqrt[4]{x^3 - x^2}}$

Soln: $y = \frac{2}{\sqrt[4]{x^3 - x^2 - x}} = 2(x^3 - x^2$

$\frac{dy}{dx} = 2\left(-\frac{1}{4}\right)(x^3 - x^2 - x)^{-\frac{5}{4}} \frac{d}{dx}(x^3 - x^2 - x$

$= -\frac{1}{2}(x^3 - x^2 - x)^{-\frac{5}{4}}(3x^2 - 2x - 1)$

$= \frac{-(3x^2 - 2x - 1)}{2(x^3 - x^2 - x)^{\frac{5}{4}}}$

$= \frac{1 + 2x - 3x^2}{2\left(\sqrt[4]{x^3 - x^2 - x}\right)^5}$

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$$\frac{d}{dx}\left(x + \frac{1}{x}\right)^{-3} = ?$$

Soln:

$$\frac{d}{dx}\left(x+\frac{1}{x}\right)^{-3} = -3\left(x+\frac{1}{x}\right)^{-4}\frac{d}{dx}\left(x+\frac{1}{x}\right)$$

$$= -3\left(x+\frac{1}{x}\right)^{-4}\left(1-\frac{1}{x^2}\right).$$

THE PRODUCT RULE

If u and v are differentiable functions of x, then

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

⇒ If we are asked to differentiate the product of a function :

ⓐ We may have to keep the first function constant and differentiate the second function

plus

ⓑ keep the second function constant and differentiate the first function

Note: If u, v and w are differentiable functions of x, then

$$\frac{d}{dx}(uvw) = uv\frac{dw}{dx} + uw\frac{dv}{dx} + vw\frac{du}{dx}$$

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Proof:

$$\frac{d}{dx}(uvw) = \frac{d}{dx}[(uv)w] = uv\frac{dw}{dx} + w\frac{d}{dx}(uv)$$

$$= uv\frac{dw}{dx} + w\left[u\frac{dv}{dx} + v\frac{du}{dx}\right]$$

$$= uv\frac{dw}{dx} + uw\frac{dv}{dx} + vw\frac{du}{dx}$$

WORKED EXAMPLES

1. Differentiate $y = (x^2+1)(x^4-2x)$

Soln: $y = (x^2+1)(x^4-2x)$

$$\frac{dy}{dx} = (x^2+1)\frac{d}{dx}(x^4-2x) + (x^4-2x)\frac{d}{dx}(x^2+1$$

$$= (x^2+1)(4x^3-2) + (x^4-2x)(2x)$$

$$= 4x^5 - 2x^2 + 4x^3 - 2 + 2x^5 - 4x^2$$

$$= 6x^5 + 4x^3 - 6x^2 - 2$$

2. $\frac{d}{dx}\left(x+\frac{1}{x}\right)^{-3} = ?$

Soln: $\frac{d}{dx}\left(x+\frac{1}{x}\right)^{-3} = -3\left(x+\frac{1}{x}\right)^{-4}\frac{d}{dx}\left(x+\frac{1}{x}\right)$

$$= -3\left(x+\frac{1}{x}\right)^{-4}\left(1-\frac{1}{x^2}\right)$$

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3. Find $\frac{dy}{dx}$ if $y = x^5(3x^3 - 2x + 1)^3$

Soln: $y = x^5(3x^3 - 2x + 1)^3$

$$\frac{dy}{dx} = x^5 \frac{d}{dx}(3x^3 - 2x + 1)^3 + (3x^3 - 2x + 1)^3 \frac{d}{dx}(x^5)$$

$$= x^5 \cdot 3(3x^3 - 2x + 1)^2 \frac{d}{dx}(3x^3 - 2x + 1) + (3x^3 - 2x + 1)^3 (5x^4)$$

$$= 3x^5(3x^3 - 2x + 1)^2(9x^2 - 2) + 5x^4(3x^3 - 2x + 1)^3$$

$$= x^4(3x^3 - 2x + 1)^2 [3x(9x^2 - 2) + 5(3x^3 - 2x + 1)]$$

$$= x^4(3x^3 - 2x + 1)^2(27x^3 - 6x + 15x^3 - 10x + 5)$$

$$= x^4(3x^3 - 2x + 1)^2(42x^3 - 16x + 5)$$

4. Calculate $\frac{dy}{dx}$ if $y = (2x^4 - 1)^4(x^8 - 3x^2)^7$

Soln: $y = (2x^4 - 1)^4(x^8 - 3x^2)^7$

$$\frac{dy}{dx} = (2x^4 - 1)^4 \frac{d}{dx}(x^8 - 3x^2)^7 + (x^8 - 3x^2)^7 \frac{d}{dx}(2x^4 - 1)^4$$

$$\frac{dy}{dx} = (2x^4 - 1)^4 \cdot 7(x^8 - 3x^2)^6 \frac{d}{dx}(x^8 - 3x^2) + (x^8 - 3x^2)^7 \cdot 4(2x^4 - 1)^3 \frac{d}{dx}(2x^4 - 1)$$

$$\frac{dy}{dx} = 7(2x^4 - 1)^4(x^8 - 3x^2)^6(8x^7 - 6x) + 4(2x^4 - 1)^3(x^8 - 3x^2)^7(8x^3)$$

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$= (2x^4-1)^3(x^8-3x^2)^6[7(2x^4-1)(8x^7-6x) + 32x^3(x^8-3x^2)$

$= (2x^4-1)^3(x^8-3x^2)^6(112x^{11}-84x^5-56x^7+42x+32x^{11}-$

$= (2x^4-1)^3(x^8-3x^2)^6(144x^{11}-56x^7-180x^5+$

$= (2x^4-1)^3(x^8-3x^2)^6 \cdot x(144x^{10}-56x^6-180x^4+$

$= x(2x^4-1)^3(x^8-3x^2)^6(144x^{10}-56x^6-180x^4+$

5. Find $\frac{dy}{dx}$ if $y = (2x^3-1)^5\sqrt{x^2-1}$

Soln: $y = (2x^3-1)^5\sqrt{x-1}$

$\frac{dy}{dx} = (2x^3-1)^5\frac{d}{dx}(x-1)^{\frac{1}{2}} + (x-1)^{\frac{1}{2}}\frac{d}{dx}(2x^3-1)^5$

$= (2x^3-1)^5 \cdot \frac{1}{2}(x-1)^{-\frac{1}{2}}\frac{d}{dx}(x-1) + (x-1)^{\frac{1}{2}} \cdot 5(2x^3-1)^4\frac{d}{dx}($

$= (2x^3-1)^5 \cdot \frac{1}{2}(x-1)^{-\frac{1}{2}}(1) + (x-1)^{\frac{1}{2}} \cdot 5(2x^3-1)^4(6$

$= \frac{1}{2}(2x^3-1)^5(x-1)^{-\frac{1}{2}} + (x-1)^{\frac{1}{2}}(2x^3-1)^4(30$

$= \frac{\frac{1}{2}(2x^3-1)^5}{(x-1)^{\frac{1}{2}}} + 30x^2(x-1)^{\frac{1}{2}}(2x^3-1)^4$

$= \frac{(2x^3-1)^5}{2\sqrt{x-1}} + 30x^2(2x^3-1)^4\sqrt{x-1}$

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6. Find $\frac{dy}{dx}$ if $y = x^2(2x-1)(6x+5)$

Soln: $y = x^2(2x-1)(6x+5)$

$$\frac{dy}{dx} = x^2(2x-1)\frac{d}{dx}(6x+5) + x^2(6x+5)\frac{d}{dx}(2x-1) + (2x-1)(6x+5)\frac{d}{dx}(x^2)$$

$$= x^2(2x-1)(6) + x^2(6x+5)(2) + (2x-1)(6x+5)(2x)$$

$$= 6x^2(2x-1) + 2x^2(6x+5) + (2x-1)(12x^2+10x)$$

$$= 12x^3 - 6x^2 + 12x^3 + 10x^2 + 24x^3 + 20x^2 - 12x^2 - 10x$$

$$= 48x^3 + 12x^2 - 10x$$

7. Given that $y = (3x-1)^2(2x^2-3)(x^3+4)^5$

Soln: $y = (3x-1)^2(2x^2-3)(x^3+4)^5$

$$\frac{dy}{dx} = (3x-1)^2(2x^2-3)\frac{d}{dx}(x^3+4)^5 + (3x-1)^2(x^3+4)^5\frac{d}{dx}(2x^2-3) + (2x^2-3)(x^3+4)^5\frac{d}{dx}(3x-1)^2$$

$$= (3x-1)^2(2x^2-3)\cdot 5(x^3+4)^4\frac{d}{dx}(x^3+4) + (3x-1)^2(x^3+4)^5(4x) + (2x^2-3)(x^3+4)^5(2)(3x-1)\frac{d}{dx}(3x-1)$$

$$= (3x-1)^2(2x^2-3)\cdot 5(x^3+4)^4(3x^2) + 4x(3x-1)^2(x^3+4)^5 + 2(2x^2-3)(x^3+4)^5(3x-1)\cdot 3$$

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$$= 15x^2(2x^2-3)(3x-1)^2(x^3+4)^4 + 4x(3x-1)^2(x^3+4)$$
$$+ 6(3x-1)(2x^2-3)(x^3+4)^5$$
$$= (3x-1)(x^3+4)^4[15x^2(2x^2-3)(3x-1) + 4x(3x-1)(x^3+4$$
$$+ 6(2x^2-3)(x^3+4)]$$

8. Calculate y' when $y = (4x^2-x)^3(x^3-2x^2)^4$

Soln.

$$y = (4x^2-x)^3(x^3-2x^2)^4(x-1)^6$$

$$\frac{dy}{dx} = (4x^2-x)^3(x^3-2x^2)^4\frac{d}{dx}(x-1)^6 + (4x^2-x)^3(x-1)^6\frac{d}{dx}(x^3-2x$$
$$+ (x^3-2x^2)^4(x-1)^6\frac{d}{dx}(4x^2$$

$$= (4x^2-x)^3(x^3-2x^2)^4\cdot 6(x-1)^5\frac{d}{dx}(x-1) + (4x^2-x)^3(x-1)^6\cdot 4(x^3-$$
$$+ (x^3-2x^2)^4(x-1)^6\cdot 3(4x^2-x)^2\cdot\frac{d}{dx}(4x^2-x)$$

$$= (4x^2-x)^3(x^3-2x^2)^4\cdot 6(x-1)^5(1) + (4x^2-x)^3(x-1)^6\cdot 4(x^3-$$
$$+ (x^3-2x^2)^4(x-1)^6\cdot 3(4x^2-x)^2\cdot(8x-1)$$

$$= 6(4x^2-x)^3(x^3-2x^2)^4(x-1)^5 + 4(4x^2-x)^3(x-1)^6(x^3-2x^2$$
$$+ 3(x^3-2x^2)^4(x-1)^6(4x^2-x)^2(8x-1)$$

$=$ . . . . . .

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8. Find $\frac{dy}{dx}$ if $y=(8x^3-x^2)^5(7x^2-3x)^2(x^5-2)^{10}$.

Sol^n

$$y=(8x^3-x^2)^5(7x^2-3x)^2(x^5-2)^{10}$$

$$\frac{dy}{dx}=(8x^3-x^2)^5(7x^2-3x)^2\frac{d}{dx}(x^5-2)^{10}+(8x^3-x^2)^5(x^5-2)^{10}\frac{d}{dx}(7x^2-3x)^2$$
$$+(7x^2-3x)^2(x^5-2)^{10}\frac{d}{dx}(8x^3-x^2)^5$$

$$\frac{dy}{dx}=(8x^3-x^2)^5(7x^2-3x)^2\cdot 10(x^5-2)^9\frac{d}{dx}(x^5-2)$$
$$+(8x^3-x^2)^5(x^5-2)^{10}\cdot 2(7x^2-3x)\frac{d}{dx}(7x^2-3x)$$
$$+(7x^2-3x)^2(x^5-2)^{10}\cdot 5(8x^3-x^2)^4\frac{d}{dx}(8x^3-x^2)$$

$$\frac{dy}{dx}=(8x^3-x^2)^5(7x^2-3x)^2\cdot 10(x^5-2)^9\cdot 5x^4$$
$$+(8x^3-x^2)^5(x^5-2)^{10}\cdot 2(7x^2-3x)\cdot(14x-3)$$
$$+(7x^2-3x)^2(x^5-2)^{10}\cdot 5(8x^3-x^2)^4(24x^2-2x)$$

$$\frac{dy}{dx}=50x^4(7x^2-3x)^2(8x^3-x^2)^5(x^5-2)^9$$
$$+2(14x-3)(7x^2-3x)(8x^3-x^2)^5(x^5-2)^{10}$$
$$+5(24x^2-2x)(7x^2-3x)^2(8x^3-x^2)^4(x^5-2)^{10}$$

$$\frac{dy}{dx}=(7x^2-3x)(8x^3-x^2)^4(x^5-2)^9\left[50x^4(7x^2-3x)(8x^3-x^2)\right.$$
$$\left.+2(14x-3)(8x^3-x^2)(x^5-2)+5(24x^2-2x)(7x^2-3x)(x^5-2)\right]$$

$$\frac{dy}{dx}=(7x^2-3x)(8x^3-x^2)^4(x^5-2)^9\left[(400x^7-50x^6)(7x^2-3x)\right.$$
$$\left.+(2x^5-4)(14x-3)(8x^3-x^2)+(5x^5-10)(24x^2-2x)(7x^2-3x)\right]$$

9. Find y' when $y=(2x^5-3x^3)^4(5x-4x^2)^3(x^6-6x^2)^7$

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Solution :

$$y = (2x^5 - 3x^3)^4 (5x - 4x^2)^3 (x^6 - 6x^2)^7$$

$$\frac{dy}{dx} = (2x^5 - 3x^3)^4 (5x - 4x^2)^3 \frac{d}{dx}(x^6 - 6x^2)^7 + (2x^5 - 3x^3)^4 (x^6 - 6x^2)^7 \frac{d}{dx}(5x - 4x^2)^3 + (5x - 4x^2)^3 (x^6 - 6x^2)^7 \frac{d}{dx}(2x^5 - 3x^3)^4$$

$$\frac{dy}{dx} = (2x^5 - 3x^3)^4 (5x - 4x^2)^3 \cdot 7(x^6 - 6x^2)^6 \frac{d}{dx}(x^6 - 6x^2) + (2x^5 - 3x^3)^4 (x^6 - 6x^2)^7 \cdot 3(5x - 4x^2)^2 \frac{d}{dx}(5x - 4x + (5x - 4x^2)^3 (x^6 - 6x^2)^7 \cdot 4(2x^5 - 3x^3)^3 \frac{d}{dx}(2x^5 - 3x$$

$$\frac{dy}{dx} = 7(2x^5 - 3x^3)^4 (5x - 4x^2)^3 (x^6 - 6x^2)^6 (6x^5 - 12x) + 3(2x^5 - 3x^3)^4 (x^6 - 6x^2)^7 (5x - 4x^2)^2 (5 - 8x) + 4(2x^5 - 3x^3)^3 (x^6 - 6x^2)^7 (5x - 4x^2)^3 (10x^4 - 9x^2)$$

$$\frac{dy}{dx} = 7(6x^5 - 12x)(2x^5 - 3x^3)^4 (5x - 4x^2)^3 (x^6 - 6x^2)^6 + 3(5 - 8x)(2x^5 - 3x^3)^4 (5x - 4x^2)^2 (x^6 - 6x^2)^7 + 4(10x^4 - 9x^2)(2x^5 - 3x^3)^3 (5x - 4x^2)^3 (x^6 - 6x^2)^7$$

$$\frac{dy}{dx} = (2x^5 - 3x^3)^3 (5x - 4x^2)^2 (x^6 - 6x^2)^6 [7(6x^5 - 12x)(2x^5 - 3x^3)(5x - 4x^2) + 3(5 - 8x)(2x^5 - 3x^3)(x + 4(10x^4 - 9x^2)(5x - 4x^2)(x^6 - 6x^2)].$$

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Note:
If f and g are differentiable at x, then

$$\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]$$

Ex 10. Find $\frac{dy}{dx}$ if $y = (4x^2-1)(7x^3+x)$

Soln:

$$\frac{dy}{dx} = \frac{d}{dx}[(4x^2-1)(7x^3+x)]$$

$$= (4x^2-1)\frac{d}{dx}(7x^3+x) + (7x^3+x)\frac{d}{dx}(4x^2-1)$$

$$= (4x^2-1)(21x^2+1) + (7x^3+x)(8x)$$

$$= 140x^4 - 9x^2 - 1$$

Ex 11. If $f(x) = (x^2+x)(2x^3-3)$, find $f'(x)$

Soln: $f'(x) = (x^2+x)\frac{d}{dx}(2x^3-3) + (2x^3-3)\frac{d}{dx}(x^2+x$

$$= (x^2+x)(6x^2) + (2x^3-3)(2x+1$$

$$= 6x^4 + 6x^3 + 4x^4 - 6x + 2x^3 - 3$$

$$= 10x^4 + 8x^3 - 6x - 3$$

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Ex 12. If $g(x) = x^{\frac{1}{5}}(x-1)^{\frac{3}{5}}$, find $g'(x)$

Soln: $g'(x) = x^{\frac{1}{5}} \cdot \frac{3}{5}(x-1)^{-\frac{2}{5}} + (x-1)^{\frac{3}{5}} \cdot \frac{1}{5} x^{-\frac{4}{5}}$

$$= \frac{3x^{\frac{1}{5}}}{5(x-1)^{\frac{2}{5}}} + \frac{1}{5}\frac{(x-1)^{\frac{3}{5}}}{x^{4/5}}$$

$=$

Ex 13 $f(x) = x^2(1-3x^3)^{\frac{1}{3}}$; $f'(x) = ?$

$$f'(x) = x^2 \frac{d}{dx}(1-3x^3)^{\frac{1}{3}} + (1-3x^3)^{\frac{1}{3}}\left(\frac{d}{dx}(x^2)\right)$$

$$= x^2\left[\frac{1}{3}(1-3x^3)^{-\frac{2}{3}}\frac{d}{dx}(1-3x^3)\right] + (1-3x^3)^{\frac{1}{3}}(\quad$$

$$= \frac{1}{3}x^2(1-3x^3)^{-\frac{2}{3}}(-9x^2) + 2x(1-3x^3$$

$$= -3x^4(1-3x^3)^{-\frac{2}{3}} + 2x(1-3x^3)^{\frac{1}{3}}$$

$$= x(1-3x^3)^{-\frac{2}{3}}[-3x^3 + 2(1-3x^3)]$$

$$= x(1-3x^3)^{-\frac{2}{3}}(2-9x^3)$$

$$= \frac{x(2-9x^3)}{(1-3x^3)^{2/3}}$$

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QUOTIENT RULE

If $f(x) = \frac{u(x)}{v(x)}$, where u and v are differentiable functions of x, then

$$f'(x) = \frac{v(x)u'(x) - u(x)v'(x)}{[v(x)]^2}$$

where $v(x) \neq 0$

Thus if u and v are differentiable funct of x and $y = \frac{u}{v}$, where $v \neq 0$, then

$$\frac{dy}{dx} = y' = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

WORKED EXAMPLES

If $y = \frac{x+1}{x-1}$, find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{(x-1)\frac{d}{dx}(x+1) - (x+1)\frac{d}{dx}(x-1)}{(x-1)^2}$$

$$= \frac{(x-1)(1) - (x+1)(1)}{(x-1)^2} = \frac{x-1-x-1}{(x-1)^2}$$

$$= \frac{-2}{(x-1)^2} = -\frac{2}{(x-1)^2}$$

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2. Find $\frac{dy}{dx}$ if $y = \frac{x^3}{x^2+1}$.

Sol^n

$$y = \frac{x^3}{x^2+1}$$

$$\frac{dy}{dx} = \frac{(x^2+1)\frac{d}{dx}(x^3) - x^3\frac{d}{dx}(x^2+1)}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{(x^2+1)(3x^2) - x^3(2x)}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{3x^4+3x^2-2x^4}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{x^4+3x^2}{(x^2+1)^2}$$

$$\therefore \frac{dy}{dx} = \frac{x^2(x^2+3)}{(x^2+1)^2}$$

3. Calculate y' when $y = \frac{2x^5-x^3}{3x^4-2x}$

Sol^n

$$y = \frac{2x^5-x^3}{3x^4-2x}$$

$$\frac{dy}{dx} = \frac{(3x^4-2x)\frac{d}{dx}(2x^5-x^3) - (2x^5-x^3)\frac{d}{dx}(3x^4-}{(3x^4-2x)^2}$$

$$\frac{dy}{dx} = \frac{(3x^4-2x)(10x^4-3x^2) - (2x^5-x^3)(12x^3-2}{(3x^4-2x)^2}$$

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$$\frac{dy}{dx} = \frac{30x^8 - 9x^6 - 20x^5 + 6x^3 - 24x^8 + 4x^5 + 12x^6 - 2x^3}{(3x^4 - 2x)^2}$$

$$\frac{dy}{dx} = \frac{6x^8 + 3x^6 - 16x^5 + 4x^3}{(3x^4 - 2x)^2}$$

$$\frac{dy}{dx} = \frac{x^3(6x^5 + 3x^3 - 16x^2 + 4)}{(3x^4 - 2x)^2}$$

2. Find $\frac{dy}{dx}$ if $y = \frac{(3x^2 - x)^4}{(x^3 - 2x^4)^3}$.

Soln

$$y = \frac{(3x^2 - x)^4}{(x^3 - 2x^4)^3}$$

$$\frac{dy}{dx} = \frac{(x^3 - 2x^4)^3 \frac{d}{dx}(3x^2 - x)^4 - (3x^2 - x)^4 \frac{d}{dx}(x^3 - 2x^4)^3}{(x^3 - 2x^4)^{3 \cdot 2}}$$

$$\frac{dy}{dx} = \frac{(x^3 - 2x^4)^3 \cdot 4(3x^2 - x)^3 \frac{d}{dx}(3x^2 - x) - (3x^2 - x)^4 \cdot 3(x^3 - 2x^4)^2 \frac{d}{dx}(x^3 - 2x^4)}{(x^3 - 2x^4)^6}$$

$$\frac{dy}{dx} = \frac{4(x^3 - 2x^4)^3(3x^2 - x)^3(6x - 1) - 3(x^3 - 2x^4)^2(3x^2 - x)^4(3x^2 - 8x^3)}{(x^3 - 2x^4)^6}$$

$$= \frac{(x^3 - 2x^4)^2(3x^2 - x)^3\left[4(6x - 1)(x^3 - 2x^4) - 3(3x^2 - 8x^3)(3x^2 - x)\right]}{(x^3 - 2x^4)^6}$$

$$= \frac{(3x^2 - x)^3}{(x^3 - 2x^4)^4}(24x^4 - 48x^5 - 4x^3 + 8x^4 - 27x^4 + 9x^3 + 72x^5 - 24x^4$$

$$= \frac{(3x^2 - x)^3}{(x^3 - 2x^4)^4}(24x^5 - 19x^4 + 5x^3)$$

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5. Calculate $\frac{dy}{dx}$ if $y = \left(\frac{2x-3}{5x^2+1}\right)^7$

Sol^n

$$y = \left(\frac{2x-3}{5x^2+1}\right)^7$$

$$\frac{dy}{dx} = 7\left(\frac{2x-3}{5x^2+1}\right)^6 \frac{d}{dx}\left(\frac{2x-3}{5x^2+1}\right)$$

$$\frac{dy}{dx} = 7\left(\frac{2x-3}{5x^2+1}\right)^6 \left[\frac{(5x^2+1)\frac{d}{dx}(2x-3) - (2x-3)\frac{d}{dx}(5x^2+1)}{(5x^2+1)^2}\right]$$

$$\frac{dy}{dx} = 7\left(\frac{2x-3}{5x^2+1}\right)^6 \left[\frac{(5x^2+1)(2) - (2x-3)(10x)}{(5x^2+1)^2}\right]$$

$$\frac{dy}{dx} = 7\left(\frac{2x-3}{5x^2+1}\right)^6 \left[\frac{10x^2+2-20x^2+30x}{(5x^2+1)^2}\right]$$

$$\frac{dy}{dx} = 7\left(\frac{2x-3}{5x^2+1}\right)^6 \left(\frac{2+30x-10x^2}{(5x^2+1)^2}\right)$$

$$\frac{dy}{dx} = \frac{7(2x-3)^6(2+30x-10x^2)}{(5x^2+1)^6(5x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{14(1+15x-5x^2)(2x-3)^6}{(5x^2+1)^8}$$

6. Calculate $\frac{d}{dx}\left[\frac{x^3-1}{\sqrt[5]{4x^2+3}}\right]$

Sol^n

$$\frac{d}{dx}\left[\frac{x^3-1}{\sqrt[5]{4x^2+3}}\right] = \frac{\sqrt[5]{4x^2+3}\,\frac{d}{dx}(x^3-1) - (x^3-1)\frac{d}{dx}(4x^2+3)^{1/5}}{\left(\sqrt[5]{4x^2+3}\right)^2}$$

$$= \frac{\sqrt[5]{4x^2+3}\,(3x^2) - (x^3-1)\cdot\frac{1}{5}(4x^2+3)^{-4/5}\frac{d}{dx}(4x^2+}{\sqrt[5]{(4x^2+3)^2}}$$

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$$= \frac{3x^2\sqrt[5]{4x^2+3} - \frac{1}{5}(x^3-1)(4x^2+3)^{-4/5}(8x)}{\sqrt[5]{(4x^2+3)^2}}$$

$$= \frac{3x^2\sqrt[5]{4x^2+3}}{\sqrt[5]{(4x^2+3)^2}} - \frac{8x(x^3-1)}{5} \cdot \frac{1}{\sqrt[5]{(4x^2+3)^2}} \cdot \frac{1}{(4x^2+3)^{4/5}}$$

$$= \frac{3x^2}{\sqrt[5]{(4x^2+3)}} - \frac{8x(x^3-1)}{5} \cdot \frac{1}{\sqrt[5]{(4x^2+3)^2}} \cdot \frac{1}{\sqrt[5]{(4x^2+3)^4}}$$

$$= \frac{3x^2}{\sqrt[5]{4x^2+3}} - \frac{8x(x^3-1)}{5\sqrt[5]{(4x^2+3)^6}}$$

7. Find y' when $y = \dfrac{\sqrt{x-1}}{x^3}$

Soln

$$y = \frac{\sqrt{x-1}}{x^3}$$

$$\frac{dy}{dx} = \frac{x^3 \frac{d}{dx}(x-1)^{1/2} - (x-1)^{1/2}\frac{d}{dx}(x^3)}{(x^3)^2}$$

$$\frac{dy}{dx} = \frac{x^3 \cdot \frac{1}{2}(x-1)^{-1/2}\frac{d}{dx}(x-1) - (x-1)^{1/2}(3x^2)}{x^6}$$

$$\frac{dy}{dx} = \frac{\frac{1}{2}x^3(x-1)^{-1/2}(1) - (x-1)^{1/2}(3x^2)}{x^6}$$

$$\frac{dy}{dx} = \frac{2(x-1)^{1/2}}{2(x-1)^{1/2}}\left[\frac{\frac{1}{2}x^3(x-1)^{-1/2} - 3x^2(x-1)^{1/2}}{x^6}\right]$$

$$\frac{dy}{dx} = \frac{x^3 - 6x^2(x-1)}{2x^6(x-1)^{1/2}} = \frac{x^3 - 6x^3 + 6x^2}{2x^6\sqrt{x-1}}$$

$$\frac{dy}{dx} = \frac{6x^2 - 5x^3}{2x^6\sqrt{x-1}} = \frac{x^2(6-5x)}{2x^6\sqrt{x-1}}$$

$$\frac{dy}{dx} = \frac{6-5x}{2x^4\sqrt{x-1}}$$

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9. If $g(t) = \frac{t^3}{1-t^4}$, find $g'(t)$

Soln: $g'(t) = \frac{(1-t^4)(3t^2) - t^3(-4}{(1-t^4)^2}$

$= \frac{3t^2 - 3t^6 + 4t^6}{(1-t^4)^2}$

$= \frac{3t^2 + t^6}{(1-t^4)^2}$

$= \frac{t^2(3+t^4)}{(1-t^4)^2}$

10. Given that $f(x) = \frac{2x+3}{x^2+1}$, find

Soln:

$f'(x) = \frac{(x^2+1)(2) - (2x+3)(2x}{(x^2+1)^2}$

$= \frac{2x^2+2 - 4x^2 - 6x}{(x^2+1)^2}$

$= \frac{-2x^2 - 6x + 2}{(x^2+1)^2}$

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Exx

Differentiate the following functions:

1. $f(x) = x^2(3x^2-1)$
2. $y = 6x^4(x^3+2x)$
3. $f(x) = (1-x^2)(1+x^2)$
4. $g(x) = \frac{1}{x^3}(x^2-x)$
5. $y = \left(\frac{2}{x}+1\right)\left(\frac{1}{x^2}-3\right)$
6. $g(x) = (3x^4-1)(5x^2-6x+11)$
7. $s = (t^2+1)(2t^2-t+3)$
8. $y = (1-\sqrt{x})(2+\sqrt{x})$
9. $f(x) = \frac{2}{x^3}$
10. $y = \frac{x}{x+1}$
11. $y = \frac{2x}{x+1}$
12. $f(x) = \frac{x-3}{2x+5}$
13. $g(x) = \frac{5x+1}{x^2-1}$
14. $y = \frac{x^2}{x^3+4}$
15. $y = \frac{x^2-x+1}{x^2+2x-3}$
16. $g(x) = \frac{7x^4+11}{x-2}$
17. $f(x) = (1+\frac{1}{x})(1+\frac{1}{x^2})$
18. $f(x) = (9x^8-8x^9)(x+\frac{1}{x})$
19. $g(x) = \frac{2x^2+1}{x+5}$
20. $f(x) = \frac{x^4}{4} - \frac{x^3}{3} + \frac{x^2}{2}$
21. $y = \frac{4x^3+1}{x^3-1}$
22. $y = \left(\frac{x}{1+x}\right)\left(\frac{2-x}{3}\right)$
23. $f(x) = (2x-3)\left(\frac{2x+3}{x}\right)$
24. $f(x) = (7x^3-4x^2+2)^{1/4}$

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DERIVATIVE OF NEGATIVE POWERS

For each negative integer n, $f(x) = x^n$ has the derivative

$$f'(x) = nx^{n-1}$$

In particular

$$f(x) = x^{-1}; \quad f'(x) = (-1)x^{-2} = -x^{-2}$$

$$f(x) = x^{-2}; \quad f'(x) = -2x^{-3} = -\frac{2}{x^3}$$

$$f(x) = x^{-3}; \quad f'(x) = -3x^{-4} = -\frac{3}{x^4}$$

Example:

If $f(x) = \frac{5}{x^2} - \frac{6}{x}$, find $f'(x)$ and f'

Soln:

$$f(x) = 5x^{-2} - 6x^{-1}$$

$$f'(x) = 5(-2)x^{-3} - 6(-1)x^{-2}$$

$$= -\frac{10}{x^3} + \frac{6}{x^2}$$

$$f'\left(\frac{1}{2}\right) = -\frac{10}{\left(\frac{1}{2}\right)^3} + \frac{6}{\left(\frac{1}{2}\right)^2}$$

$$= -80 + 24 = -56$$

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* Note Because $y = \frac{1}{x^n}$ can be expressed as $y = x^{-n}$ problems where quotient rule is applicable, can be inverted to products so that you apply product rule. e, it is a matter of choice or preference.

WORKED EXAMPLES

Differentiate :

(i) $(x^2-2)(x+3)^{-2}$ as a product

(ii) $\frac{x^2-2}{(x+3)^2}$ as a quotient

Soln

Let $y = (x^2-2)(x+3)^{-2}$

$$\frac{dy}{dx} = (x^2-2)\frac{d}{dx}(x+3)^{-2} + (x+3)^{-2}\frac{d}{dx}(x^2-2)$$ [Product

$$\frac{dy}{dx} = (x^2-2)\cdot -2(x+3)^{-3}\frac{d}{dx}(x+3) + (x+3)^{-2}(2x)$$

$$\frac{dy}{dx} = -\frac{2(x^2-2)}{(x+3)^3}(1) + \frac{2x}{(x+3)^2}$$

$$\frac{dy}{dx} = \frac{-2x^2+4}{(x+3)^3} + \frac{2x}{(x+3)^2}$$

$$\frac{dy}{dx} = \frac{-2x^2+4+2x(x+3)}{(x+3)^3}$$

$$\frac{dy}{dx} = \frac{-2x^2+4+2x^2+6x}{(x+3)^3}$$

$$\frac{dy}{dx} = \frac{4+6x}{(x+3)^3} = \frac{2(2+3x)}{(x+3)^3}$$

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(ii)

$$y = \frac{x^2-2}{(x+3)^2}$$

$$\frac{dy}{dx} = \frac{(x+3)^2 \frac{d}{dx}(x^2-2) - (x^2-2)\frac{d}{dx}(x+3)^2}{\{(x+3)^2\}^2}$$

$$\frac{dy}{dx} = \frac{(x+3)^2(2x) - (x^2-2)\cdot 2(x+3)\frac{d}{dx}(x+3)}{(x+3)^4}$$

$$\frac{dy}{dx} = \frac{2x(x+3)^2 - 2(x+3)(x^2-2)(1)}{(x+3)^4}$$

$$\frac{dy}{dx} = \frac{(x+3)\{2x(x+3) - 2(x^2-2)}{(x+3)^4}$$

$$\frac{dy}{dx} = \frac{2x^2+6x-2x^2+4}{(x+3)^3} = \frac{6x+4}{(x+3)^3} = \frac{2(2+}{(x+}$$

NB: On comparing $\frac{dy}{dx}$ for (i) and (ii), you will notice th final results are the same. Hence,

$$y = \frac{x^2-2}{(x+3)^2} = (x^2-2)(x+3)^{-2}.$$

2. Differentiate : (i) $(x-1)^3(x^3-1)^{-1}$ as a product

(ii) $\frac{(x-1)^3}{(x^3-1)}$ as a quotient.

Sol^n

(i) $y = (x-1)^3(x^3-1)^{-1}$

$$\frac{dy}{dx} = (x-1)^3 \frac{d}{dx}(x^3-1)^{-1} + (x^3-1)^{-1}\frac{d}{dx}(x-1)^3$$

$$\frac{dy}{dx} = (x-1)^3\cdot(-1)(x^3-1)^{-2}\frac{d}{dx}(x^3-1) + (x^3-1)^{-1}\cdot 3(x-1)^2$$

$$\frac{dy}{dx} = -(x-1)^3(x^3-1)^{-2}(3x^2) + 3(x^3-1)^{-1}(x-1)^2(1)$$

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$$\frac{dy}{dx} = \frac{-3x^2(x-1)^3}{(x^3-1)^2} + \frac{3(x-1)^2}{(x^3-1)}$$

$$\frac{dy}{dx} = \frac{-3x^2(x-1)^3 + 3(x-1)^2(x^3-1)}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{(x-1)^2\left[-3x^2(x-1) + 3(x^3-1)\right]}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{(x-1)^2(-3x^3+3x^2+3x^3-3)}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{(x-1)^2(3x^2-3)}{(x^3-1)^2} = \frac{3(x^2-1)(x-1)^2}{(x^3-1)^2} = \frac{3(x+1)(x-1)^3}{(x^3-1)^2}$$

$$y = \frac{(x-1)^3}{x^3-1}$$

$$\frac{dy}{dx} = \frac{(x^3-1)\frac{d}{dx}(x-1)^3 - (x-1)^3\frac{d}{dx}(x^3-1)}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{(x^3-1)\cdot 3(x-1)^2\frac{d}{dx}(x-1) - (x-1)^3(3x^2)}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{3(x^3-1)(x-1)^2(1) - 3x^2(x-1)^3}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{3(x-1)^2\left[(x^3-1) - x^2(x-1)\right]}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{3(x-1)^2(x^3-1-x^3+x^2)}{(x^3-1)^2}$$

$$\frac{dy}{dx} = \frac{3(x-1)^2(x^2-1)}{(x^3-1)^2} = \frac{3(x-1)^2(x-1)(x+1)}{(x^3-1)^2}$$

$$\therefore \frac{dy}{dx} = \frac{3(x+1)(x-1)^3}{(x^3-1)^2}$$

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3. Differentiate (ii) $(x^2-1)^3 x^{-4}$ as a product

(iii) $\frac{(x^2-1)^3}{x^4}$ as a quotient.

Sol^n

(ii) $y = (x^2-1)^3 x^{-4}$

$$\frac{dy}{dx} = (x^2-1)^3 \frac{d}{dx}(x^{-4}) + x^{-4} \frac{d}{dx}(x^2-1)^3$$

$$\frac{dy}{dx} = (x^2-1)^3 \cdot (-4)x^{-5} + x^{-4} \cdot 3(x^2-1)^2 \frac{d}{dx}(x^2-1)$$

$$\frac{dy}{dx} = \frac{-4(x^2-1)^3}{x^5} + 3x^{-4}(x^2-1)^2(2x)$$

$$\frac{dy}{dx} = \frac{-4(x^2-1)^3}{x^5} + \frac{6x(x^2-1)^2}{x^4}$$

$$\frac{dy}{dx} = \frac{-4(x^2-1)^3 + 6x^2(x^2-1)^2}{x^5}$$

$$\frac{dy}{dx} = \frac{(x^2-1)^2[-4(x^2-1) + 6x^2]}{x^5}$$

$$\frac{dy}{dx} = \frac{(x^2-1)^2(-4x^2+4+6x^2)}{x^5} = \frac{(x^2-1)^2(2x^2+4)}{x^5}$$

$$\frac{dy}{dx} = \frac{2(x^2+2)(x^2-1)^2}{x^5}$$

(iii) $y = \frac{(x^2-1)^3}{x^4}$

$$\frac{dy}{dx} = \frac{x^4 \frac{d}{dx}(x^2-1)^3 - (x^2-1)^3 \frac{d}{dx}(x^4)}{(x^4)^2}$$

$$\frac{dy}{dx} = \frac{x^4 \cdot 3(x^2-1)^2 \frac{d}{dx}(x^2-1) - (x^2-1)^3 \cdot 4x^3}{x^8}$$

$$\frac{dy}{dx} = \frac{3x^4(x^2-1)^2(2x) - 4x^3(x^2-1)^3}{x^8}$$

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$$\frac{dy}{dx} = \frac{6x^5(x^2-1)^2 - 4x^3(x^2-1)^3}{x^8}$$

$$\frac{dy}{dx} = \frac{2x^3(x^2-1)^2\left[3x^2 - 2(x^2-1)\right]}{x^8}$$

$$\frac{dy}{dx} = \frac{2(x^2-1)^2(3x^2 - 2x^2 + 2)}{x^5}$$

$$\frac{dy}{dx} = \frac{2(x^2-1)^2(x^2 + 2)}{x^5}$$

HIGHER - ORDER DERIVATIVES

A given function $f(x)$ can be differentiated twice, thrice etc. If the given function $f(x)$ is differentiated twice, we get its second derivative. It is denoted by $f''(x)$ or

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right).$$

When $f(x)$ is differentiated thrice, we get its third eriviative. It is denoted by $f'''(x)$ or

$$\frac{d^3y}{dx^2} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d}{dx}\left\{\frac{d}{dx}\left(\frac{dy}{dx}\right)\right\} \quad \text{etc.}$$

WORKED EXAMPLES

.. Given the function $y = x^2 - x - \frac{1}{5x^6}$, find $\frac{d^2y}{dx^2}$.

Sol^n

$$y = x^2 - x - \frac{1}{5x^6}$$

$$y = x^2 - x - \frac{1}{5}x^{-6}$$

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$$\frac{dy}{dx} = 2x - 1 - \frac{1}{5}(-6)x^{-7}$$

$$\frac{dy}{dx} = 2x - 1 + \frac{6}{5}x^{-7}$$

But $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(2x - 1 + \frac{6}{5}x^{-7}\right)$

$$\frac{d^2y}{dx^2} = 2 + \frac{6}{5}(-7)x^{-8}$$

$$\frac{d^2y}{dx^2} = 2 - \frac{42}{5}x^{-8} = 2 - \frac{42}{5x^8}$$

2. Find y'' when $y = 3\sqrt{x} - \frac{2}{\sqrt{x}}$.

Sol^n

$$y = 3\sqrt{x} - \frac{2}{\sqrt{x}}$$

$$y = 3x^{\frac{1}{2}} - 2x^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = 3 \cdot \frac{1}{2}x^{-\frac{1}{2}} - 2\left(-\frac{1}{2}\right)x^{-\frac{3}{2}}$$

$$\frac{dy}{dx} = \frac{3}{2}x^{-\frac{1}{2}} + x^{-\frac{3}{2}}$$

But $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{3}{2}x^{-\frac{1}{2}} + \right.$

$$\frac{d^2y}{dx^2} = \frac{3}{2}\left(-\frac{1}{2}\right)x^{-\frac{3}{2}} + \left(-\frac{3}{2}\right)x^{-\frac{5}{2}}$$

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$\frac{d^2y}{dx^2} = -\frac{3}{4}x^{-5/2} - \frac{3}{2}x^{-5/2}$

$\frac{d^2y}{dx^2} = \frac{-3}{4x^{5/2}} - \frac{3}{2x^{5/2}}$

$\frac{d^2y}{dx^2} = \frac{-3}{4\sqrt{x^5}} - \frac{3}{2\sqrt{x^5}}$

3. If $y = (3x^2 - 2x)^3$, find $\frac{d^2y}{dx^2}$.

Soln

$y = (3x^2 - 2x)^3$

$\frac{dy}{dx} = 3(3x^2-2x)^2 \cdot \frac{d}{dx}(3x^2 - 2x)$

$\frac{dy}{dx} = 3(3x^2-2x)^2(6x - 2)$

But $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left[3(3x^2-2x)^2(6x-2)\right]$

$\frac{d^2y}{dx^2} = 3(3x^2-2x)^2 \frac{d}{dx}(6x-2) + 3(6x-2)\frac{d}{dx}(3x^2-2x)^2$

$\frac{d^2y}{dx^2} = 3(3x^2-2x)^2(6) + 3(6x-2)\cdot 2(3x^2-2x)\frac{d}{dx}(3x^2-2x$

$\frac{d^2y}{dx^2} = 18(3x^2-2x)^2 + 6(6x-2)(3x^2-2x)(6x-2)$

$\frac{d^2y}{dx^2} = 6(3x^2-2x)\left[3(3x^2-2x) + (6x-2)^2\right]$

$\frac{d^2y}{dx^2} = 6(3x^2-2x)(9x^2-6x + 36x^2 - 24x + 4)$

$\frac{d^2y}{dx^2} = 6(3x^2-2x)(45x^2 - 30x + 4)$

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4. Find $\frac{d^2y}{dx^2}$ when $y = (x^2-1)(x^3+2)$.

Sol^n

$$y = (x^2-1)(x^3+2)$$

$$\frac{dy}{dx} = (x^2-1)\frac{d}{dx}(x^3+2) + (x^3+2)\frac{d}{dx}(x^2-1)$$

$$\frac{dy}{dx} = (x^2-1)(3x^2) + (x^3+2)(2x)$$

$$\frac{dy}{dx} = 3x^4 - 3x^2 + 2x^4 + 4x = 5x^4 - 3x^2 + 4x$$

But $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(5x^4 - 3x^2 + 4x)$

$$\frac{d^2y}{dx^2} = 20x^4 - 6x + 4$$

5. Calculate y'' when $y = \frac{1}{\sqrt[3]{x-1}}$.

Sol^n

$$y = \frac{1}{\sqrt[3]{x-1}} = \frac{1}{(x-1)^{1/3}} = (x-1)^{-1/3}$$

$$\frac{dy}{dx} = -\frac{1}{3}(x-1)^{-4/3}\frac{d}{dx}(x-1)$$

$$\frac{dy}{dx} = -\frac{1}{3}(x-1)^{-4/3}(1) = -\frac{1}{3}(x-1)^{-4/3}$$

But $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left[-\frac{1}{3}(x-1)^{-4/3}\right]$

$$\frac{d^2y}{dx^2} = \left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)(x-1)^{-7/3}\frac{d}{dx}(x-1)$$

$$\frac{d^2y}{dx^2} = \frac{4}{9}(x-1)^{-7/3}(1)$$

$$\therefore \frac{d^2y}{dx^2} = \frac{4}{9(x-1)^{7/3}} = \frac{4}{9\sqrt[3]{(x-1)^7}}$$

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Ex

ind the indicated derivative for each of the
llowing functions :

1. $y = (3x+1)^4$, $\frac{d^2y}{dx^2}$

2. $y = (x^2+1)^6$, $y''(5)$

3. $\frac{d^2}{dt^2}\left(\frac{t^4}{2t^3-1}\right)$

4. $\frac{d^2}{dx^2}\left[\frac{x}{x-2}\right]$

5. $f(x) = \frac{1}{\sqrt{x^2+1}}$, $f''(x)$

6. $f(x) = (x-3)^5$, $f'''(x)$, $f'''(2)$

7. $y = (1-2x)^4$, $\frac{d^3y}{dx^3}$

8. $f(x) = \frac{1}{\sqrt{x}}$, $f^4(x)$

9. $y = \frac{2x}{\sqrt[3]{x-1}}$, $\frac{d^2y}{dx^2}$

10. $f(x) = \frac{1}{x^2}$, $f''(x)$

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IMPLICIT DIFFERENTIATION

This is a method for finding the deriv of a function without first solving the eq explicitly for y in terms of x.

All the equations for which we have calculate the derivatives have been functions that can be expressed in the form $y = f(x)$.

This form, in which the equation is solved for variable (usually y) is called an EXPLICIT form

Examples are

$y = x^2 + 2x$; $y = \sqrt{1-x}$; $y = \frac{x^2}{x^2+1}$

Consider the following functions:

$x^2 + y^2 = 4$; $\sqrt{x} + \sqrt{y} = 1$; $x^2y - 2y^3 =$

In such cases, the process to calculate the derivatives is called IMPLICIT DIFFERENTIATION

Ex 1. If $y^3 = x^2$, find $\frac{dy}{dx}$

Soln: $3y^2 \frac{dy}{dx} = 2x$

$$\frac{dy}{dx} = \frac{2x}{3y^2}$$

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Ex 2. Given that $x^2+y^2=4$, find $\frac{dy}{dx}$

Soln: $2x\frac{dx}{dx}+2y\frac{dy}{dx}=0$

$$2x+2y\frac{dy}{dx}=0$$

$$2y\frac{dy}{dx}=-2x$$

$$\frac{dy}{dx}=-\frac{x}{y}$$

Ex 3. If $x^3+x^2y^2-2y=0$, find $\frac{dy}{dx}$

Soln: $3x^2+\left(x^2\left[2y\frac{dy}{dx}\right]+y^2(2x)\right)-2\frac{dy}{dx}=0$

$$3x^2+2x^2y\frac{dy}{dx}+2xy^2-2\frac{dy}{dx}=0$$

$$(2x^2y-2)\frac{dy}{dx}=-3x^2-2xy^2$$

$$\frac{dy}{dx}=\frac{-3x^2-2xy^2}{2x^2y-2}$$

$$=\frac{3x^2+2xy^2}{2}$$

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6. If $x^2 - 9y^2 = 16$, find $\frac{dy}{dt}$

Soln. $\frac{d}{dt}[x^2] - 9\frac{d}{dt}[y^2] = \frac{d}{dt}[16]$

$$2x\frac{dx}{dt} - 18y\frac{dy}{dt} = 0$$

$$-18y\frac{dy}{dt} = -2x\frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{x}{9y}\frac{dx}{dt}$$

7. If $xy + y^2 = 1$, find $\frac{dy}{dx}$

Soln: $x\frac{dy}{dx} + y + 2y\frac{dy}{dx} = 0$

$$(x+2y)\frac{dy}{dx} = -y$$

$$\frac{dy}{dx} = -\frac{y}{x+2y}$$

8. If $x^3 - y^3 = 1$, find $\frac{dy}{dx}$

Soln: $x^3 - y^3 = 1$

$$3x^2 - 3y^2\frac{dy}{dx} = 0$$

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$$2y\frac{dy}{dx} = -3x^2$$

$$\therefore \frac{dy}{dx} = -\frac{3x^2}{2y}$$

1. Given that $x^2 - 4xy + 3y^2 = 7y + 5x$, find $\frac{dy}{dx}$.

Soln

$$x^2 - 4xy + 3y^2 = 7y + 5x$$

$$\frac{d}{dx}(x^2) - 4\frac{d}{dx}(xy) + 3\frac{d}{dx}(y^2) = 7\frac{d}{dx}(y) + 5\frac{d}{dx}(x)$$

$$2x - 4\left[x\frac{dy}{dx} + y\frac{dx}{dx}\right] + 3\cdot 2y\frac{dy}{dx} = 7\frac{dy}{dx} + 5$$

$$2x - 4x\frac{dy}{dx} - 4y + 6y\frac{dy}{dx} = 7\frac{dy}{dx} + 5$$

$$-4x\frac{dy}{dx} + 6y\frac{dy}{dx} - 7\frac{dy}{dx} = 5 - 2x + 4y$$

$$(6y - 4x - 7)\frac{dy}{dx} = 5 - 2x + 4y$$

$$\therefore \frac{dy}{dx} = \frac{5 - 2x + 4y}{6y - 4x - 7}$$

2. Find $\frac{dy}{dx}$ when $y^3 - x^3 = 4xy$.

Soln

$$y^3 - x^3 = 4xy$$

$$\frac{d}{dx}(y^3) - \frac{d}{dx}(x^3) = 4\frac{d}{dx}(xy)$$

$$3y^2\frac{dy}{dx} - 3x^2 = 4\left[x\frac{dy}{dx} + y\frac{dx}{dx}\right]$$

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Exx

Using implicit differentiation, find $\frac{dy}{dx}$ of the eff:

1. $x - 3y = -1$
2. $x - 3y^2 + 1 = 0$
3. $y^2 = 2x - 1$
4. $y^2 = 2x^2 - 1$
5. $x^2 - 4y^2 = 4$
6. $9x^2 - 16y^2 = 144$
7. $x^3 + y^3 = 8$
8. $4x^3 - 8y^3 = 32$
9. $x^2y^2 - 4y^3 + 3x = 0$
10. $x^2y^2 - 4y^3 + 3x^2 = 0$

$2xy - 4xy^2 + x^2y = 0$

$5xy - y^4 = x$

$\sqrt{xy} = 1$

14. $\sqrt{x} + \sqrt{y} = 5$
15. $(x+y)^2 = 1$
16. $\frac{1}{x} + \frac{1}{y} = 1$
17. $x^3 - 3xy^2 + y^3 = 1$
18. $y^3 - x^2 = 4$
19. $x^3 + y^3 - 3xy = 0$
20. $4x^2 + 9y^2 = 36$
21. $\sin(x+y) = xy$
22. $\tan xy = xy$
23. $x^4 + 4x^3y + y^4 = 1$
24. $x^2 - 2xy + 4y^2 =$
25. $y^2 + xy - x^2 =$

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THE CHAIN RULE

Suppose that y is a differentiable fu
of u $(y = f(u))$ and
u in turn is a differentiable funct
x $(u = g(x))$.

Then y is a composite function o

$$y = f(u)$$
$$= f\{g(x)\}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Ex If $y = 2u$ and $u = 3x$,
then $y = 6x$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$
$$= (2)(3$$
$$= 6$$

$y = 6x$

$\frac{dy}{dx} = 6$

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x If $y = u^3$ and $u = x^2$, $\frac{dy}{dx} = ?$

Soln: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= (3u^2)(2x)$

$= 3(x^2)^2(2x)$

$= 6x^5$

Ex Find $\frac{dy}{dx}$ given that $y = \frac{u-1}{u+1}$ and $u = x^2$

Soln:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= \left[\frac{(u+1)(1) - (u-1)(1)}{(u+1)^2}\right](2x)$$

$$= \left[\frac{2}{(u+1)^2}\right](2x)$$

$$= \frac{4x}{(x^2+1)^2}$$

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The formula $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ can be ex
to more variables. For example, if x
depends on s, then

$$\frac{dy}{ds} = \frac{dy}{du} \cdot \frac{du}{dx} \cdot \frac{dx}{ds}$$

If, in addition, s depends on t,

$$\frac{dy}{dt} = \frac{dy}{du} \cdot \frac{du}{dx} \cdot \frac{dx}{ds} \cdot \frac{ds}{dt}$$

And so we notice that each new d
adds a new link to the chain.

Ex. Find $\frac{dy}{ds}$ given that $y = 3u + 1$,

$u = \frac{1}{x^2}$ and $x = 1 - s$

Soln: $\frac{dy}{ds} = \frac{dy}{du} \cdot \frac{du}{dx} \cdot \frac{dx}{ds}$

$= (3)(-2x^{-3})(-1) = 6(1-s)^{-3}$

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< Find $\frac{dy}{dt}$ at $t = 9$ given that

$$y = \frac{u+2}{u-1}, \quad u = (3s-7)^2, \quad s = \sqrt{t}$$

$2(3s-7)$

ɔln : $\frac{dy}{du} = -\frac{3}{(u-1)^2}$; $\frac{du}{ds} = 6(3s-7)$

$$\frac{ds}{dt} = \frac{1}{2\sqrt{t}}$$

At $t = 9$, we have $s = 3$ and $u = 4$

$$\frac{dy}{du} = -\frac{3}{(4-1)^2} = -\frac{1}{3} \; ; \; \frac{du}{ds} = 6(9-7) = 12$$

$$\frac{ds}{dt} = \frac{1}{2\sqrt{9}} = \frac{1}{6}$$

ns at $t = 9$,

$$\frac{dy}{dt} = \frac{dy}{du} \cdot \frac{du}{ds} \cdot \frac{ds}{dt}$$

$$= \left(-\frac{1}{3}\right)(12)\left(\frac{1}{6}\right) = -\frac{2}{3}$$

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Exx

Find $\frac{dy}{dx}$ at $x=0$ of the ff:

1. $y = \frac{1}{1+u^2}$, $u = 2x+1$

2. $y = u + \frac{1}{u}$, $u = (3x+\ )^4$

3. $y = \frac{2u}{1-4u}$, $u = (5x^2+1)^4$

4. $y = u^3 - u + 1$, $u = \frac{1-x}{1+x}$

Exx Find $\frac{dy}{dt}$

1. $y = \frac{1-7u}{1+u^2}$, $u = 1+x^2$, $x = 2t-5$

2. $y = 1+u^2$, $u = \frac{1-7x}{1+x^2}$, $x = 5t+$

Find $\frac{dy}{dx}$ at $x=2$

$= (s+3)^2$, $s = \sqrt{t-3}$, $t = x^2$

$= \frac{1+s}{1-s}$, $s = t - \frac{1}{t}$, $t = 5t+2$

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PARAMETRIC EQUATIONS

In some cases, it is more convenient to represent a function by expressing x and y separately in terms of a third independent variable called a parameter eg. $y = \cos 2t$, $x = \sin t$

The two equations are parametric equations and the parameter is t

Eg1. Given $y = \cos 2t$, $x = \sin t$, find $\frac{dy}{dx}$

Soln: $\frac{dy}{dt} = -2\sin 2t$; $\frac{dx}{dt} = \cos t$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = (-2\sin 2t)\left(\frac{1}{\cos t}\right)$$

$$= -4\sin t \cos t \cdot \frac{1}{\cos t}$$

$$= -4\sin t$$

2. If $x = \frac{2-3t}{1+t}$, $y = \frac{3+2t}{1+t}$, find $\frac{dy}{dx}$

Soln: $x = \frac{2-3t}{1+t} \Rightarrow \frac{dx}{dt} = \frac{(1+t)(-3) - (2-3t)(1)}{(1+t)^2}$

$y = \frac{3+2t}{1+t} \Rightarrow \frac{dy}{dt} = \frac{(1+t)(2) - (3+2t)(1)}{(1+t)^2}$

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$$\frac{dx}{dt} = \frac{-5}{(1+t)^2} \; ; \; \frac{dy}{dt} = \frac{-1}{(1+t)^2}$$

But $$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = \frac{-1}{(1+t)^2} \cdot \frac{(1+t)^2}{-5}$$

$$= \frac{1}{5}$$

3. If $x = a(\cos\theta + \theta\sin\theta)$ and $y = a(\sin\theta - \theta\cos\theta)$, find $\frac{dy}{dx}$

Soln: $x = a(\cos\theta + \theta\sin\theta)$

$$\frac{dx}{d\theta} = a(-\sin\theta + \theta\cos\theta + \sin\theta) = a$$

$$\frac{dy}{d\theta} = a(\cos\theta + \theta\sin\theta - (\cos\theta) = a\theta s$$

But $$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{a\theta\sin\theta \cdot 1}{a\theta\cos\theta}$$

$$= \tan\theta$$

Exx

1. If $x = 3\cos\theta - \cos^3\theta$ and $y = 3\sin\theta - \sin^3\theta$, express $\frac{dy}{dx}$ in terms of θ

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ERNATIVES OF EXPONENTIAL FUNCTIONS

Let $y = e^x$. Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = e^x \cdot \frac{d}{dx}(x)$$

$$= e^x(1)$$

$$= e^x$$

j. $f(x) = e^{x^2}$; $f'(x) = ?$

n: $f'(x) = e^{x^2}\left[\frac{d}{dx}(x^2)\right]$

$$= e^{x^2}(2x)$$

$$= 2xe^{x^2}$$

Let $f(x) = e^{5x^8}$: $f'(x) = ?$

Soln: $f'(x) = e^{5x^8}\left[\frac{d}{dx}(5x^8)\right]$

$$= e^{5x^8}[40x^7]$$

$$= 40x^7 e^{5x^8}$$

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4 Let $y = \frac{e^x}{e^x+1}$ Find $\frac{dy}{dx}$

Soln: $$\frac{dy}{dx} = \frac{(e^x+1)\frac{d}{dx}(e^x) - e^x\frac{d}{dx}(e^x+1)}{(e^x+1)^2}$$

$$= \frac{(e^x+1)e^x - e^x(e^x)}{(e^x+1)^2}$$

$$= \frac{e^{2x}+e^x-e^{2x}}{(e^x+1)^2}$$

$$= \frac{e^x}{(e^x+1)^2}$$

5 Let $y = xe^x$. Find $\frac{dy}{dx}$

Soln $$\frac{dy}{dx} = x[e^x] + e^x[1]$$

$$= xe^x + e^x$$

$$= e^x(x+1)$$

Let $y = x^2e^{-x}$. Find $\frac{dy}{dx}$

Soln $$\frac{dy}{dx} = x^2(-e^{-x}) + e^{-x}(2x)$$

$$= -x^2e^{-x} + 2xe^{-x}$$

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9 Find $\frac{dy}{dx}$ when $y = e^{3(1-x^2)^8}$

Soln: $y = e^{3(1-x^2)^8}$

$$\frac{dy}{dx} = e^{3(1-x^2)^8} \frac{d}{dx}[3(1-x^2)^8]$$

$$= e^{3(1-x^2)^8} \cdot 3(8)(1-x^2)^7 \cdot \frac{d}{dx}(1-x^2)$$

$$= e^{3(1-x^2)^8} \cdot 24(1-x^2)^7(-2x)$$

$$= -48x(1-x^2)^7 e^{3(1-x^2)^8}$$

10 Calculate $\frac{d}{dx}[e^{2x}\cos 5x^3]$

Soln:

$$\frac{d}{dx}[e^{2x}\cos 5x^3] = e^{2x}\frac{d}{dx}(\cos 5x^3) + \cos 5x^3 \frac{d}{dx}(e^{2x})$$

$$= e^{2x}(-\sin 5x^3)\frac{d}{dx}(5x^3) + (\cos 5x^3)e^{2x}\frac{d}{dx}(2x)$$

$$= -e^{2x}\sin 5x^3(15x^2) + \cos 5x^3 e^{2x}(2)$$

$$= -15x^2 e^{2x}\sin 5x^3 + 2e^{2x}\cos 5x^3$$

$$= e^{2x}(2\cos 5x^3 - 15x^2 \sin 5x^3)$$

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NATIVES OF LOGARITHMIC FUNCTIONS

$y = \ln x$ then

$$\frac{dy}{dx} = \frac{1}{x}\frac{d}{dx}(x)$$

1. Let $y = \ln(4x^2+1)$. Find $\frac{dy}{dx}$

Soln:

$$\frac{dy}{dx} = \frac{1}{4x^2+1}\frac{d}{dx}(4x^2+1)$$

$$= \frac{1}{4x^2+1}(8x)$$

$$= \frac{8x}{4x^2+1}$$

2. If $f(x) = \ln(1-x^3)$. Find $f'(-2)$

Soln:

$$f(x) = \ln(1-x^3)$$

$$f'(x) = \frac{1}{1-x^3}\frac{d}{dx}(1-x^3)$$

$$= \frac{1}{1-x^3}(-3x^2)$$

$$= -\frac{3x^2}{1-x^3}$$

$$f'(-2) = -\frac{3(-2)^2}{1-(-2)^3} = -\frac{4}{3}.$$

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Ex 3. Find $\frac{dy}{dx}$ given that $y = \ln(4x-1)$

Soln:

$$y = \ln(4x-1)$$

$$\frac{dy}{dx} = \frac{1}{4x-1} \cdot \frac{d}{dx}(4x-1)$$

$$= \frac{1}{4x-1}(4)$$

$$= \frac{4}{4x-1}$$

Ex 4. Let $y = (\ln x)^3$. Find $\frac{dy}{dx}$

Soln:

$$y = (\ln x)^3$$

$$\frac{dy}{dx} = 3(\ln x)^2 \cdot \frac{d}{dx}(\ln x)$$

$$= 3(\ln x)^2 \cdot \frac{1}{x}$$

$$= \frac{3}{x}(\ln x)^2$$

Ex 5. Let $y = \ln\left(\frac{x-1}{x+1}\right)$. Find $\frac{dy}{dx}$

Soln:

$$\frac{dy}{dx} = \frac{1}{\left(\frac{x-1}{x+1}\right)} \cdot \frac{d}{dx}\left(\frac{x-1}{x+1}\right)$$

$$= \frac{x+1}{x-1}\left[\frac{(x+1)\frac{d}{dx}(x-1) - (x-1)\frac{d}{dx}(x+1)}{(x+1)^2}\right]$$

$$= \frac{x+1}{x-1} \cdot \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2}$$

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$$\frac{x+1}{x-1}\,\frac{(x+1)-(x-1)}{(x+1)^2} = \frac{x+1}{x-1}\cdot\frac{x+1-x+1}{(x+1)^2}$$

$$= \frac{1}{x-1}\cdot\frac{2}{x+1}$$

$$= \frac{2}{(x-1)(x+1)}$$

6 If $y = x^2 \ln x$, find $\frac{dy}{dx}$

Soln:

$$y = x^2 \ln x$$

$$\frac{dy}{dx} = x^2 \frac{d}{dx}(\ln x) + \ln x \frac{d}{dx}(x^2)$$

$$= x^2 \cdot \frac{1}{x} + \ln x\,(2x)$$

$$= x + 2x \ln x$$

$$= x(1 + 2\ln x)$$

Let $y = \ln(\ln x)$. Find $\frac{dy}{dx}$

Soln:

$$y = \ln(\ln x)$$

$$\frac{dy}{dx} = \frac{1}{\ln x}\cdot\frac{d(\ln x)}{dx}$$

$$= \frac{1}{\ln x}\cdot\frac{1}{x}$$

$$= \frac{1}{x \ln x}$$

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DERIVATIVES OF TRIGONOMETRIC FUNC

$$\frac{d}{dx}(\sin u) = \cos u \cdot \frac{du}{dx}$$

$$\frac{d}{dx}(\cos u) = -\sin u \cdot \frac{du}{dx}$$

$$\frac{d}{dx}(\tan u) = \sec^2 u \cdot \frac{du}{dx}$$

$$\frac{d}{dx}(\csc u) = -\csc u \cot u \frac{du}{dx}$$

$$\frac{d}{dx}(\sec u) = \sec u \tan u \frac{du}{dx}$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u \frac{du}{dx}$$

Ex 1

$$\frac{d}{dx}(\cos 2x) = -\sin 2x \frac{d}{dx}(2x)$$

$$= -\sin 2x (2)$$

$$= -2 \sin 2x$$

Ex 2. $$\frac{d}{dx}(\sin 3x) = \cos 3x \cdot \frac{d}{dx}(3x)$$

$$= \cos 3x (3)$$

$$= 3 \cos 3x$$

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Ex 6. Calculate $\frac{d}{dx}(\cos^2 x)$

Soln: $\cos^2 x = (\cos x)^2$

$\frac{d}{dx}(\cos^2 x) = 2(\cos x)\cdot\frac{d}{dx}(\cos x)$

$= 2\cos x(-\sin x)$

$= -2\cos x \sin x$

$= -\sin 2x$

Ex 7. Find $\frac{d}{dx}(\sqrt{\sin x})$

Soln: $\sqrt{\sin x} = (\sin x)^{\frac{1}{2}}$

$\frac{d}{dx}(\sqrt{\sin x}) = \frac{1}{2}(\sin x)^{-\frac{1}{2}}\cdot\frac{d}{dx}(\sin x)$

$= \frac{1}{2\sqrt{\sin x}}(\cos x)$

$= \frac{\cos x}{2\sqrt{\sin x}}$

Ex 8. Show that $\frac{d}{dx}(\tan x) = \sec^2 x$

Soln: $\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$

$= \frac{\cos x \frac{d}{dx}(\sin x) - \sin x}{(\cos x)^2}$

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$$= \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} = \sec^2 x$$

9. Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$

Soln:

$$\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = \frac{d}{dx}(\cos x)^{-1}$$

$$\frac{d}{dx}(\cos x)^{-1} = (-1)(\cos x)^{-2} \cdot \frac{d}{dx}(\cos x)$$

$$= -(\cos x)^{-2}(-\sin x)$$

$$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$$

$$= \sec x \tan x$$

10. Calculate $\frac{d}{dx}(x \sin x)$

Soln:

$$\frac{d}{dx}(x \sin x) = x \cdot \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(x$$

$$= x \cos x + \sin x (1)$$

$$= x \cos x + \sin x$$

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LEVEL 2

Ex 1. Find $\frac{d}{dx}\left(\frac{\sin x}{x}\right)$

Soln: $\frac{d}{dx}\left(\frac{\sin x}{x}\right) = \frac{x\frac{d}{dx}(\sin x) - \sin x}{x^2}$

$= \frac{x\cos x - \sin x(1)}{x^2}$

$= \frac{x\cos x - \sin x}{x^2}$

Ex 2. Find $\frac{dy}{dx}$ if $y = \sin^3(5x+4)$

Soln: $y = \sin^3(5x+4) = [\sin(5x+4)$

$\frac{dy}{dx} = 3[\sin(5x+4)]^2 \frac{d}{dx}[\sin(5x+$

$= 3\sin^2(5x+4)\cos(5x+4)\cdot\frac{d}{dx}$

$= 3\sin^2(5x+4)\cos(5x+4)(5)$

$= 15\sin^2(5x+4)\cos(5x+4)$

Ex 3. Find $\frac{d}{dx}\left(2\tan\left(\frac{x}{2}\right) - 5\right)$

Soln: $\frac{d}{dx}\left(2\tan\left(\frac{x}{2}\right) - 5\right) = 2\sec^2\left(\frac{x}{2}\right)\cdot$

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$= 2\sec^2\left(\frac{x}{2}\right)\left(\frac{1}{2}\right)$

$= \sec^2\frac{x}{2}$

4. Find $\frac{d}{dx}(\cot^2 x)$

Soln: $\cot^2 x = [\cot x]^2$

$\frac{d}{dx}(\cot^2 x) = \frac{d}{dx}[\cot x]^2$

$= 2\cot x \cdot \frac{d}{dx}(\cot x)$

$= 2\cot x(-\csc^2 x)$

$= -2\cot x \operatorname{cosec}^2 x$

$= -2\cot x \csc^2 x$

Find $\frac{d}{dx}[\csc(5x-3)]$

$\frac{d}{dx}[\csc(5x-3)] = -\csc(5x-3)\cot(5x-3)\frac{d}{dx}(5x-$

$= -\csc(5x-3)\cot(5x-3)(5)$

$= -5\csc(5x-3)\cot(5x-3)$

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8. Find $\frac{dy}{dx}$ if $y = \cos^5(4x^3+2x^2+8)$

Soln: $y = \cos^5(4x^3+2x^2+8) = [\cos(4x^3+2x^2+8)]^5$

$\frac{dy}{dx} = 5[\cos(4x^3+2x^2+8)]^4 \frac{d}{dx}[\cos(4x^3+2x^2+8)]$

$= 5\cos^4(4x^3+2x^2+8)[-\sin(4x^3+2x^2+8)]\frac{d}{dx}(4x^3+2x^2+8)$

$= -5\cos^4(4x^3+2x^2+8)\sin(4x^3+2x^2+8)(12x^2+4x)$

$= -5(12x^2+4x)\cos^4(4x^3+2x^2+8)\sin(4x^3+2x^2+8)$

Ex 9. Find $\frac{dy}{dx}$ if $y = \tan^6(4x^5+2x^3+6x+2)$

Soln: $y = \tan^6(4x^5+2x^3+6x+2) = [\tan(4x^5+2x^3+6x+2)]^6$

$\frac{dy}{dx} = 6[\tan(4x^5+2x^3+6x+2)]^5 \frac{d}{dx}[\tan(4x^5+2x^3+6x+2)]$

$= 6\tan^5(4x^5+2x^3+6x+2)\sec^2(4x^5+2x^3+6x+2)\frac{d}{dx}(4x^5+2x^3+6x+2)$

$6\tan^5(4x^5+2x^3+6x+2)\sec^2(4x^5+2x^3+6x+2)(20x^4+6x^2+6)$

$(20x^4+6x^2+6)\tan^5(4x^5+2x^3+6x+2)\sec^2(4x^5+2x^3+6x+2)$

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e 2 Calculate $\frac{d}{dx}\left(\sqrt{\cos^5 4x^3}\right)$

n :

$$\frac{d}{dx}\left(\sqrt{\cos^5 4x^3}\right) = \frac{d}{dx}\left(\cos 4x^3\right)^{\frac{5}{2}}$$

$$= \frac{5}{2}\left(\cos 4x^3\right)^{\frac{3}{2}} \frac{d}{dx}\left(\cos 4x^3\right)$$

$$= \frac{5}{2}\left(\cos 4x^3\right)^{\frac{3}{2}}\left(-\sin 4x^3\right)\frac{d}{dx}\left(4x^3\right)$$

$$= -\frac{5}{2}\left(\cos 4x^3\right)^{\frac{3}{2}} \sin 4x^3 \left(12x^2\right)$$

$$= -\frac{5}{2}\left(12x^2\right)\left(\cos 4x^3\right)^{\frac{3}{2}} \sin 4x^3$$

$$= -30x^2 \sin 4x^3 \sqrt{\cos^3 4x^3}$$

x 3. Find $\frac{dy}{dx}$ given that $y = (3x^5 - 1)\sin 7x^2$

ion :

$$y = (3x^5 - 1)\sin 7x^2$$

$$\frac{dy}{dx} = (3x^5 - 1)\frac{d}{dx}(\sin 7x^2) + \sin 7x^2 \frac{d}{dx}(3x^5 - 1)$$

$$= (3x^5 - 1)(\cos 7x^2)\frac{d}{dx}(7x^2) + \sin 7x^2 (15x^4)$$

$$= (3x^5 - 1)\cos 7x^2 (14x) + 15x^4 \sin 7x^2$$

$$= 14x(3x^5 - 1)\cos 7x^2 + 15x^4 \sin 7x^2$$

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INVERSE TRIGONOMETRIC FUNCTION

$$\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}} \cdot \frac{d(x)}{dx}; \quad \frac{d}{dx}(\cos^{-1}x) =$$

$$\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2} \cdot \frac{d(x)}{dx}$$

Ex 1. Find $\frac{dy}{dx}$ given that $y = \sin^{-1}x$

Soln:

$$y = \sin^{-1}x$$

$$\Rightarrow x = \sin y$$

$$1 = \cos y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

But $\sin^2 y + \cos^2 y = 1$

$$\cos^2 y = 1 - \sin^2 y = 1 - x^2$$

$$\cos y = \sqrt{1-}$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

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2. Find $\frac{dy}{dx}$ if $y = \cos^{-1} x$

Soln: $y = \cos^{-1} x \Rightarrow x = \cos y$

$$1 = -\sin y \frac{dy}{dx}$$

$$\frac{dy}{dx} = -\frac{1}{\sin y}$$

But $\sin^2 y + \cos^2 y = 1$

$$\sin^2 y = 1 - \cos^2 y = 1 - x^2$$

$$\sin y = \sqrt{1-x^2}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

3 Find $\frac{dy}{dx}$ if $y = \tan^{-1} x$

Soln: $y = \tan^{-1} x$

$$\Rightarrow x = \tan y$$

$$1 = \sec^2 y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

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$y = \sin^{-1} 8x^2$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(8x^2)^2}} \cdot \frac{d}{dx}(8x^2)$$

$$= \frac{1}{\sqrt{1-64x^4}}(16x)$$

$$= \frac{16x}{\sqrt{1-64x^4}}$$

6 If $y = \tan^{-1}(3x+5)$, find $\frac{dy}{dx}$

$$y = \tan^{-1}(3x+5)$$

$$\frac{dy}{dx} = \frac{1}{1+(3x+5)^2} \frac{d}{dx}(3x+5)$$

$$= \frac{1}{1+(9x^2+30x+25)}(3)$$

$$= \frac{3}{1+9x^2+30x+25}$$

$$= \frac{3}{9x^2+30x+26}$$

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Ex 7 Find $\frac{dy}{dx}$ if $y = \tan^{-1}\left(\frac{x}{1-x}\right)$

Soln: $y = \tan^{-1}\left(\frac{x}{1-x}\right)$

$$\frac{dy}{dx} = \frac{1}{1+\left(\frac{x}{1-x}\right)^2} \cdot \frac{d}{dx}\left(\frac{x}{1-x}\right)$$

$$= \left[\frac{1}{1+\frac{x^2}{1-2x+x^2}}\right] \frac{(1-x)\frac{d}{dx}(x) - x\left(\frac{d}{dx}(1-x)\right)}{(1-x)^2}$$

$$= \left(\frac{1-2x+2x^2}{1-2x+x^2}\right) \frac{(1-x)(1) - x(-1)}{(1-x)^2}$$

$$= \frac{(1-2x+2x^2)(1-x+x)}{(1-x)^2(1-x)^2}$$

$$= \frac{1-2x+2x^2}{(1-x)^4}$$

Find $\frac{dy}{dx}$ if $y = x \sin^{-1} x^2$

Soln: $y = x \sin^{-1} x^2$

$$\frac{dy}{dx} = x \frac{d}{dx}(\sin^{-1} x^2) + \sin^{-1} x^2 \frac{d}{dx}(x)$$

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$$= x \cdot \frac{1}{\sqrt{1-(x^2)^2}} \frac{d}{dx}(x^2) + \sin^{-1} x^2 (1)$$

$$= x \frac{1}{\sqrt{1-x^4}} (2x) + \sin^{-1} x^2$$

$$= \frac{2x^2}{\sqrt{1-x^4}} + \sin^{-1} x^2$$

Ex 9* Find $\frac{dy}{dx}$ given that $y = x \tan^{-1} \sqrt{x}$

Soln:

$$y = x \tan^{-1} \sqrt{x}$$

$$\frac{dy}{dx} = x \cdot \frac{d}{dx}\left[\tan^{-1}\sqrt{x}\right] + \tan^{-1}\sqrt{x} \frac{d}{dx}(x)$$

$$= x \cdot \frac{1}{1+(\sqrt{x})^2} \frac{d}{dx}(\sqrt{x}) + \tan^{-1}\sqrt{x}(1)$$

$$= \frac{x}{1+x}\left(\frac{1}{2}x^{-\frac{1}{2}}\right) + \tan^{-1}\sqrt{x}$$

$$= \frac{x^{\frac{1}{2}}}{2(1+x)} + \tan^{-1}\sqrt{x}$$

$$= \frac{\sqrt{x}}{2(1+x)} + \tan^{-1}\sqrt{x}$$

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Exx

Find $\frac{dy}{dx}$ of the following functions

1. $y = (\sin^{-1}x)^2$

2. $y = \tan^{-1}(x+1)$

3. $y = \tan^{-1}\sqrt{x}$

4. $y = \sin^{-1}x^2$

5 $y = \ln(\tan^{-1}x)$

6. $y = e^{\tan^{-1}x}$

7. $y = x\sin^{-1}2x$

$y = \frac{\tan^{-1}x}{2x}$

$y = \tan^{-1}\left(\frac{2}{x}\right)$

$y = \frac{1}{2}\sin^{-1}(3x-4x^2)$

$y = \tan^{-1}(\ln x)$

12. $y = \sin^{-1}($

13. $y = \sqrt{\tan^{-1}}$

14 $y = \tan(\tan$

15 $y = \tan^{-1}($

16 $y = \sqrt{\sin^{-1}}$

17 $y = \ln(\sin$

18 $y = (\tan^{-1}$

19. $y = \sin^{-1}$

20 $y = \cos^{-1}$

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